

Question Number	Answer	Additional guidance	Mark
4(a)(i)	The only correct answer is C (P and R) <i>A is not correct because both P and R would shorten</i> <i>B is not correct because both P and R would shorten</i> <i>D is not correct because both P and R would shorten</i>		(1)

Question Number	Answer	Additional guidance	Mark
4(a)(ii)	A calculation with the following answer. 1.8 (: 1)	accept $\pm 1\text{mm}$ i.e. 1.78 to 1.82 (: 1) ignore ratio as fraction ignore 2:1 max 2dp.	(1)

Question Number	Answer	Additional guidance	Mark
4(a)(iii)	myosin (1)	Mark first word ONLY ignore tropomyosin ACCEPT phonetic spelling	(1)

Question Number	Answer	Additional guidance	Mark
4(b)	A description that includes four of the following points: • calcium ions {bind / attach} to the troponin (1) • causing troponin to change shape (1) • {moving / pulling / pushing} tropomyosin (1) • (therefore) exposes myosin binding site (1) • allowing myosin head to {bind / form crossbridge} (1)	Describe the role ACCEPT causes conformational change in troponin ignore confirmation change ACCEPT tropomyosin is removed (from binding site) ignore released ACCEPT myosin binding site is free ACCEPT {attach / join} for bind	(4)

Question Number	Answer	Additional guidance	Mark
4(c)(i)	An answer that includes the following points: • correct scale added to both axes (1) • appropriate line of best fit drawn (1)	e.g 100-200-300- (400) accept scale up to 350 Eg 0 – 60 DO NOT ACCEPT LINE OF BEST FIT THROUGH ORIGIN	(2)

Question Number	Answer	Additional guidance	Mark
4(c)(ii)	A calculation with the following step: • calculation of gradient (1)	0.104 correct answer gains full marks Accept 0.08 – 0.14 No ecf ignore fractions ignore units	(1)

